

LeaseGPT — Production-Grade Lease Q&A System

A system that accepts lease documents (PDF or scanned images), extracts and indexes their content, and allows users to ask natural language questions about the lease through a real-time chat interface. Answers are grounded exclusively in the uploaded lease document with citations.

Decisions Summary

Decision	Choice	Rationale
Agent Orchestration	LangGraph only (no CrewAI)	Simpler stack; LangGraph covers all orchestration needs for single-doc Q&A
RAG Evaluation	Ragas only (no DeepEval)	Handles batch eval in CI/CD; avoids redundant eval frameworks
Cloud LLM	AWS Bedrock (prod) + Ollama (dev)	Unified AWS billing, IAM auth, multi-model access
OCR	AWS Textract	Best fit for existing AWS stack, high accuracy for documents
Project Name	LeaseGPT	—

Tech Stack (Final, No Overlaps)

Layer	Technologies
Frontend	Next.js, TypeScript, TailwindCSS
Backend & API	Python 3.12, FastAPI, WebSockets
LLM & Orchestration	LangGraph, LlamaIndex, Pydantic, Tenacity, AWS Bedrock (prod), Ollama (dev)
Retrieval & Eval	Hybrid RAG (semantic + keyword), pgvector, Ragas
Database & Cache	PostgreSQL (pgvector), Redis
Document Processing	AWS Textract (OCR), PyMuPDF (PDF text extraction)
Cloud & Serverless	AWS S3, Step Functions, Lambda
DevOps	Docker, Kubernetes (kind/minikube), GitHub Actions
Testing	pytest (backend), Jest (frontend)

Architecture Overview

```

graph TB
    subgraph "Frontend - Next.js"
        UI["Chat UI + Upload"]
    end

    subgraph "API Layer - FastAPI"
        REST["REST API  
(upload, history, health)"]
        WS["WebSocket  
(streaming Q&A)"]
    end

    subgraph "Document Processing Pipeline - AWS"
        S3["S3 Bucket  
(lease storage)"]
        SF["Step Functions  
(orchestrator)"]
        L1["Lambda: Classify  
(PDF vs Image)"]
        L2["Lambda: Extract  
(Textract / PyMuPDF)"]
        L3["Lambda: Chunk & Embed  
(LlamaIndex)"]
    end

    subgraph "Q&A Engine"
        LG["LangGraph Agent  
(query→retrieve→synthesize→cite)"]
        LI["LlamaIndex  
(hybrid retrieval)"]
    end

    subgraph "Data Stores"
        PG["PostgreSQL + pgvector  
(embeddings, metadata, history)"]
        RD["Redis  
(sessions, cache, rate limiting)"]
    end

    subgraph "LLM Providers"
        BR["AWS Bedrock  
(production)"]
        OL["Ollama  
(local dev)"]
    end

    UI -->|"upload"| REST
    UI -->|"chat"| WS
    REST -->|"store file"| S3
    S3 -->|"S3 event"| SF
    SF --> L1 --> L2 --> L3
    L3 -->|"store embeddings"| PG
    WS --> LG
    LG --> LI
    LI -->|"hybrid search"| PG
    LG -->|"generate"| BR
    LG -->|"generate"| OL

```

```
REST →|"session"| RD  
WS →|"cache"| RD  
L3 →|"notify complete"| RD
```

Monorepo Structure

```

leasegpt/
├── .github/
│   └── workflows/
│       ├── ci.yml                # Lint, test, build on PR
│       ├── cd-staging.yml        # Deploy to staging on merge to develop
│       └── cd-production.yml     # Deploy to prod on tag/release
├── frontend/                    # Next.js application
│   ├── src/
│   │   ├── app/                # App Router pages
│   │   │   ├── layout.tsx
│   │   │   ├── page.tsx        # Landing / upload page
│   │   │   └── chat/
│   │   │       └── [leaseId]/
│   │   │           └── page.tsx # Chat interface per lease
│   │   ├── components/
│   │   │   ├── ui/            # Reusable UI primitives
│   │   │   │   ├── Button.tsx
│   │   │   │   ├── Card.tsx
│   │   │   │   ├── Modal.tsx
│   │   │   │   └── Spinner.tsx
│   │   │   ├── FileUpload.tsx # Drag & drop + file picker
│   │   │   ├── ChatWindow.tsx # Message list + auto-scroll
│   │   │   ├── ChatInput.tsx  # Input bar + send button
│   │   │   ├── MessageBubble.tsx # Individual message w/ citations
│   │   │   ├── LeaseCard.tsx  # Lease summary card
│   │   │   ├── ProcessingStatus.tsx # Upload progress indicator
│   │   │   └── CitationHighlight.tsx # Inline citation popover
│   │   ├── hooks/
│   │   │   ├── useWebSocket.ts # WS connection management
│   │   │   ├── useLeaseUpload.ts # Upload with progress tracking
│   │   │   └── useChat.ts      # Chat state management
│   │   ├── lib/
│   │   │   ├── api.ts          # REST API client (axios/fetch)
│   │   │   ├── ws.ts           # WebSocket client wrapper
│   │   │   └── types.ts        # Shared TypeScript types
│   │   └── styles/
│   │       └── globals.css      # TailwindCSS + custom tokens
│   ├── public/
│   ├── __tests__/              # Jest tests
│   ├── next.config.ts
│   ├── tailwind.config.ts
│   ├── tsconfig.json
│   ├── package.json
│   └── Dockerfile
├── backend/                    # FastAPI application
│   ├── app/
│   │   ├── main.py             # FastAPI app factory + lifespan
│   │   ├── config.py           # Pydantic Settings (env-based config)
│   │   └── api/
│   │       ├── __init__.py
│   │       ├── routes/
│   │       │   ├── __init__.py
│   │       │   ├── upload.py   # POST /api/leases/upload
│   │       │   ├── leases.py   # GET /api/leases, GET /api/leases/{id}
│   │       └── chat.py         # WebSocket /ws/chat/{lease_id}

```

```

├── health.py          # GET /health, GET /ready
├── dependencies.py    # FastAPI dependency injection
├── middleware.py      # CORS, rate limiting, auth
├── models/
│   ├── __init__.py
│   ├── lease.py       # SQLAlchemy: Lease, LeaseChunk
│   ├── chat.py        # SQLAlchemy: ChatSession, Message
│   └── schemas.py     # Pydantic request/response models
├── services/
│   ├── __init__.py
│   ├── upload_service.py # S3 upload + Step Functions trigger
│   ├── chat_service.py  # WebSocket message handling
│   └── lease_service.py  # Lease CRUD operations
├── rag/
│   ├── __init__.py
│   ├── indexer.py       # LlamaIndex: chunking + embedding
│   ├── retriever.py     # LlamaIndex: hybrid retrieval
│   ├── agent.py         # LangGraph: Q&A agent graph
│   ├── prompts.py       # System prompts + prompt templates
│   └── llm_provider.py  # Bedrock / Ollama provider factory
├── db/
│   ├── __init__.py
│   ├── session.py       # SQLAlchemy async session factory
│   ├── migrations/     # Alembic migrations
│   └── redis.py         # Redis connection pool
├── utils/
│   ├── __init__.py
│   ├── s3.py            # S3 client wrapper
│   └── textract.py      # Textract client wrapper
├── tests/
│   ├── conftest.py      # Fixtures (test DB, mock S3, etc.)
│   ├── unit/
│   │   ├── test_indexer.py
│   │   ├── test_retriever.py
│   │   ├── test_agent.py
│   │   └── test_schemas.py
│   ├── integration/
│   │   ├── test_upload_flow.py
│   │   ├── test_chat_flow.py
│   │   └── test_websocket.py
│   └── evaluation/
│       ├── test_ragas.py # Ragas evaluation suite
│       ├── golden_qa.json # Golden Q&A pairs for eval
│       └── conftest.py
├── pyproject.toml
├── alembic.ini
├── Dockerfile
├── lambdas/           # AWS Lambda functions
│   ├── classify/
│   │   ├── handler.py  # Detect PDF vs image
│   │   ├── requirements.txt
│   │   └── Dockerfile
│   ├── extract/
│   │   ├── handler.py  # Textract OCR / PyMuPDF extraction
│   │   ├── requirements.txt
│   │   └── Dockerfile
└── chunk_embed/

```

```

├── handler.py                # LlamaIndex chunking + pgvector insert
├── requirements.txt
├── Dockerfile

├── infra/                    # Infrastructure as Code
│   ├── docker/
│   │   ├── docker-compose.yml    # Local dev: all services
│   │   ├── docker-compose.test.yml # CI: services for integration tests
│   │   └── .env.example
│   ├── k8s/
│   │   ├── base/
│   │   │   ├── namespace.yml
│   │   │   ├── frontend-deployment.yml
│   │   │   ├── backend-deployment.yml
│   │   │   ├── postgres-statefulset.yml
│   │   │   ├── redis-deployment.yml
│   │   │   ├── ingress.yml
│   │   │   └── kustomization.yml
│   │   ├── overlays/
│   │   │   ├── dev/              # kind/minikube overrides
│   │   │   ├── staging/
│   │   │   └── production/
│   │   └── secrets/              # Sealed secrets templates
│   └── aws/
│       ├── step-functions.json    # State machine definition
│       ├── lambda-iam-role.json   # Lambda execution role
│       └── s3-event-config.json   # S3 → Step Functions trigger

├── scripts/
│   ├── setup-local.sh            # One-command local setup
│   ├── seed-db.sh                # Seed dev database
│   └── run-eval.sh                # Run Ragas evaluation suite

├── docs/
│   ├── architecture.md
│   ├── api-spec.md
│   ├── deployment-guide.md
│   └── adr/                      # Architecture Decision Records
│       ├── 001-langgraph-over-crewai.md
│       ├── 002-hybrid-rag-strategy.md
│       └── 003-bedrock-llm-provider.md

├── .env.example
├── .gitignore
├── README.md
└── Makefile                      # Common commands (make dev, make test, etc.)

```

Proposed Changes — Detailed Component Breakdown

Phase 1: Foundation (Infrastructure + Data Layer)

1.1 Database Schema — PostgreSQL + pgvector

```

-- Enable extensions
CREATE EXTENSION IF NOT EXISTS "uuid-osspl";
CREATE EXTENSION IF NOT EXISTS "vector";

-- Leases table
CREATE TABLE leases (
  id          UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  filename    VARCHAR(500) NOT NULL,
  s3_key      VARCHAR(1000) NOT NULL,
  file_type   VARCHAR(10) NOT NULL,          -- 'pdf' | 'image'
  status      VARCHAR(20) NOT NULL DEFAULT 'uploading',
                                          -- uploading → processing → ready → error
  page_count  INT,
  extracted_text TEXT,                      -- full raw text (for keyword search)
  metadata    JSONB DEFAULT '{}',          -- arbitrary metadata
  created_at  TIMESTAMPTZ DEFAULT NOW(),
  updated_at  TIMESTAMPTZ DEFAULT NOW()
);

-- Lease chunks (for RAG)
CREATE TABLE lease_chunks (
  id          UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  lease_id    UUID NOT NULL REFERENCES leases(id) ON DELETE CASCADE,
  chunk_index INT NOT NULL,
  content     TEXT NOT NULL,                -- chunk text
  embedding   VECTOR(768) NOT NULL,         -- nomic-embed-text (768d, free)
  page_number INT,
  section_title VARCHAR(500),
  metadata    JSONB DEFAULT '{}',
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

-- Hybrid search indexes
CREATE INDEX idx_chunks_embedding ON lease_chunks
  USING ivfflat (embedding vector_cosine_ops) WITH (lists = 100);
CREATE INDEX idx_chunks_content_tsvector ON lease_chunks
  USING gin (to_tsvector('english', content));
CREATE INDEX idx_chunks_lease_id ON lease_chunks(lease_id);

-- Chat sessions
CREATE TABLE chat_sessions (
  id          UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  lease_id    UUID NOT NULL REFERENCES leases(id) ON DELETE CASCADE,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

-- Chat messages
CREATE TABLE chat_messages (
  id          UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  session_id  UUID NOT NULL REFERENCES chat_sessions(id) ON DELETE CASCADE,
  role        VARCHAR(10) NOT NULL,         -- 'user' | 'assistant'
  content     TEXT NOT NULL,
  citations    JSONB DEFAULT '[]',          -- [{chunk_id, page, snippet}]
  token_count INT,
  latency_ms  INT,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

```



```
);
CREATE INDEX idx_messages_session ON chat_messages(session_id, created_at);
```

1.2 Docker Compose — Local Development

All services orchestrated for local dev with hot-reload:

Service	Port	Notes
frontend	3000	Next.js dev server with hot-reload
backend	8000	FastAPI with uvicorn <code>--reload</code>
postgres	5432	PostgreSQL 16 + pgvector extension
redis	6379	Redis 7 for caching/sessions
ollama	11434	Local LLM (auto-pulls model on first run)
localstack	4566	Mock AWS services (S3, Step Functions) for local dev

1.3 Configuration — backend/app/config.py

Pydantic Settings model with environment-based config:

```

class Settings(BaseSettings):
    # App
    app_name: str = "LeaseGPT"
    environment: Literal["local", "staging", "production"] = "local"
    debug: bool = False

    # Database
    database_url: str = "postgresql+asyncpg:// ..."
    redis_url: str = "redis://localhost:6379/0"

    # AWS
    aws_region: str = "us-east-1"
    s3_bucket: str = "leasegpt-documents"
    step_functions_arn: str = ""
    textract_enabled: bool = True

    # LLM
    llm_provider: Literal["bedrock", "ollama"] = "ollama"
    bedrock_model_id: str = "anthropic.claude-3-5-sonnet-20241022-v2:0"
    ollama_base_url: str = "http://localhost:11434"
    ollama_model: str = "llama3.1:8b"

    # Embedding
    embedding_provider: Literal["bedrock", "ollama"] = "ollama"
    embedding_model: str = "nomic-embed-text"          # free via Ollama
    embedding_dimension: int = 768                     # nomic-embed-text output dim

    # RAG
    chunk_size: int = 512
    chunk_overlap: int = 50
    top_k: int = 5
    hybrid_alpha: float = 0.7 # 0.0 = keyword only, 1.0 = semantic only

```

Phase 2: Document Processing Pipeline

2.1 Upload Flow — `backend/app/api/routes/upload.py`

```

POST /api/leases/upload
├─ Validate file (PDF/image, size < 20MB)
├─ Generate UUID + S3 key
├─ Upload to S3 (multipart for large files)
├─ Create lease record in DB (status: "processing")
├─ Trigger Step Functions execution
└─ Return { lease_id, status: "processing" }

```

2.2 Step Functions State Machine

```

stateDiagram-v2
    [*] → Classify
    Classify → ExtractText: PDF (text-based)
    Classify → OCRExtract: Image or scanned PDF
    ExtractText → ChunkAndEmbed
    OCRExtract → ChunkAndEmbed
    ChunkAndEmbed → NotifyComplete
    NotifyComplete → [*]

    Classify → HandleError: Error
    ExtractText → HandleError: Error
    OCRExtract → HandleError: Error
    ChunkAndEmbed → HandleError: Error
    HandleError → [*]

```

Lambda 1 — Classify (`lambdas/classify/handler.py`):

- Download file from S3
- Detect if PDF has extractable text or is scanned/image
- Return `{ file_type: "text_pdf" | "scanned_pdf" | "image" }`

Lambda 2 — Extract (`lambdas/extract/handler.py`):

- **Text PDF** → PyMuPDF: extract text per page
- **Image / Scanned PDF** → AWS Textract: async document analysis, extract text blocks with bounding boxes
- Store raw extracted text back to the `leases` table
- Return `{ text, page_count, pages: [{page_num, text}] }`

Lambda 3 — Chunk & Embed (`lambdas/chunk_embed/handler.py`):

- Uses LlamaIndex `SentenceSplitter` for semantic chunking
- Generates embeddings via Bedrock Titan Embeddings (or Ollama in dev)
- Batch inserts into `lease_chunks` with pgvector
- Updates lease status to `"ready"`
- Publishes completion event to Redis pub/sub (so the frontend can update)

2.3 Processing Status — Real-Time Updates

The frontend polls `GET /api/leases/{id}` or subscribes to a Redis pub/sub channel via WebSocket for real-time status updates during processing.

Phase 3: RAG & Q&A Engine

3.1 Hybrid Retrieval — `backend/app/rag/retriever.py`

Two-pronged retrieval strategy combining semantic and keyword search:

User Query

- └ Semantic Search (pgvector cosine similarity)
 - └ SELECT * FROM lease_chunks
 - WHERE lease_id = \$1
 - ORDER BY embedding $\iff$ \$query_embedding
 - LIMIT \$top_k
- └ Keyword Search (PostgreSQL full-text search)
 - └ SELECT * FROM lease_chunks
 - WHERE lease_id = \$1
 - AND to_tsvector('english', content) @@ plainto_tsquery('english', \$query)
 - ORDER BY ts_rank(...) DESC
 - LIMIT \$top_k
- └ Reciprocal Rank Fusion (RRF)
 - └ Merge and re-rank results → top K chunks

The `hybrid_alpha` config parameter controls the weighting between semantic and keyword results during RRF fusion.

3.2 LangGraph Agent — `backend/app/rag/agent.py`

Stateful graph for the Q&A workflow:

```
graph LR
  A["analyze_query"] --> B["retrieve_context"]
  B --> C["check_relevance"]
  C -->|relevant| D["generate_answer"]
  C -->|not relevant| E["refine_query"]
  E --> B
  D --> F["format_citations"]
  F --> G["stream_response"]
```

Nodes:

Node	Responsibility
<code>analyze_query</code>	Classify query intent, extract key terms, detect if it's a lease-related question
<code>retrieve_context</code>	Call hybrid retriever, get top-K chunks scoped to the specific lease
<code>check_relevance</code>	LLM-as-judge: are retrieved chunks sufficient to answer? If not, refine and retry (max 2 retries)
<code>generate_answer</code>	LLM generates answer grounded in retrieved chunks, with inline citations
<code>format_citations</code>	Structure citations as <code>[Section X, Page Y]</code> references
<code>stream_response</code>	Token-by-token streaming via WebSocket

State schema (Pydantic):

```
class QASState(BaseModel):
    lease_id: UUID
    query: str
    query_analysis: Optional[QueryAnalysis] = None
    retrieved_chunks: list[ChunkResult] = []
    relevance_check: Optional[RelevanceResult] = None
    answer: Optional[str] = None
    citations: list[Citation] = []
    retry_count: int = 0
    max_retries: int = 2
```

3.3 LLM Provider Factory — `backend/app/rag/llm_provider.py`

Abstracts Bedrock vs Ollama behind a common interface using LlamaIndex's LLM abstraction:

```
def get_llm(settings: Settings) → BaseLLM:
    if settings.llm_provider == "bedrock":
        return Bedrock(model=settings.bedrock_model_id, ...)
    else:
        return Ollama(model=settings.ollama_model, base_url=settings.ollama_base_url)
```

Tenacity retry decorator wraps all LLM calls with exponential backoff:

```
@retry(
    stop=stop_after_attempt(3),
    wait=wait_exponential(multiplier=1, min=2, max=30),
    retry=retry_if_exception_type((RateLimitError, TimeoutError)),
)
async def llm_call_with_retry(llm, prompt, **kwargs):
    ...
```

3.4 Prompt Engineering — `backend/app/rag/prompts.py`

System prompt enforces lease-only grounding:

You are LeaseGPT, a lease document analyst. You answer questions ONLY based on the provided lease document excerpts. You MUST:

1. Only use information from the provided context chunks
2. Cite specific sections/pages for every claim: [Page X, Section Y]
3. If the lease does not contain relevant information, say:

"I couldn't find information about this in your lease document."
4. Never fabricate lease terms or provide legal advice
5. Quote exact lease language when possible

Phase 4: WebSocket Chat — Real-Time Streaming

4.1 WebSocket Endpoint — `backend/app/api/routes/chat.py`

WS `/ws/chat/{lease_id}`

Client → Server messages:

```
{ "type": "query", "content": "Can I have pets?", "session_id": "... " }
```

Server → Client messages:

```
{ "type": "token", "content": "Based" }           # streaming tokens
{ "type": "token", "content": " on" }
{ "type": "citations", "data": [ ... ] }          # after answer complete
{ "type": "done", "message_id": "... " }          # end of response
{ "type": "error", "message": "... " }           # error occurred
```

4.2 Redis Integration

Use Case	Redis Feature
Response caching	Cache full responses keyed by <code>{lease_id}:{query_hash}</code> with 1h TTL
Session state	Store active WebSocket session data
Rate limiting	Sliding window rate limiter (e.g., 20 queries/minute per session)
Processing notifications	Pub/sub channel for document processing status updates

Phase 5: Frontend — Next.js

5.1 Pages

Route	Component	Description
<code>/</code>	<code>page.tsx</code>	Landing page with upload area
<code>/chat/[leaseId]</code>	<code>page.tsx</code>	Chat interface for a specific lease

5.2 Key Components

FileUpload.tsx — Drag-and-drop upload zone:

- Accepts PDF and image files (JPEG, PNG, TIFF)
- Shows upload progress bar
- Validates file size (< 20MB) client-side
- Transitions to processing status view after upload

ChatWindow.tsx — Chat interface:

- Message list with auto-scroll
- User messages (right-aligned) and assistant messages (left-aligned)
- Streaming text animation for assistant responses
- Inline citation chips that expand to show source text and page number

CitationHighlight.tsx — Citation popover:

- Clickable [Page X] badges in assistant messages
- Popover shows the exact lease text that was cited
- Links to the relevant page/section context

ProcessingStatus.tsx — Document processing tracker:

- Step-by-step progress: Upload → Extract → Index → Ready
- Animated transitions between states
- Auto-redirects to chat when processing completes

5.3 WebSocket Hook — **useWebSocket.ts**

```
function useWebSocket(leaseId: string) {
  // Manages connection lifecycle (connect, reconnect, disconnect)
  // Handles message types (token, citations, done, error)
  // Accumulates streaming tokens into complete messages
  // Exponential backoff reconnection on disconnect
}
```

Phase 6: Kubernetes Deployment

6.1 Service Architecture

```

graph TB
    subgraph "Kubernetes Cluster"
        ING["Ingress Controller  
(nginx)"]

        subgraph "Frontend Pod(s)"
            FE["Next.js"]
        replicas: 2"]
        end

        subgraph "Backend Pod(s)"
            BE["FastAPI"]
        replicas: 2-4  
(HPA)"]
        end

        subgraph "Data Layer"
            PG["PostgreSQL  
StatefulSet  
(pgvector)"]
            RD["Redis  
Deployment"]
        end
        end

        ING -->|"/api, /ws"| BE
        ING -->|"/*"| FE
        BE --> PG
        BE --> RD

```

6.2 Kustomize Overlays

Overlay	Replicas	Resources	LLM Provider	Notes
dev (kind/minikube)	1 each	Low (256Mi-512Mi)	Ollama	LocalStack for AWS mocks
staging	2 each	Medium (512Mi-1Gi)	Bedrock	Real AWS services
production	2-4 (HPA)	High (1-2Gi)	Bedrock	HPA, PDB, resource quotas

Phase 7: CI/CD — GitHub Actions

7.1 CI Pipeline (on PR)


```

ci.yml
├─ Lint & Format
│   ├── Backend: ruff, mypy
│   └─ Frontend: eslint, prettier
├─ Unit Tests
│   ├── Backend: pytest (unit/)
│   └─ Frontend: jest
├─ Integration Tests
│   ├── Spin up docker-compose.test.yml
│   ├── Run pytest (integration/)
│   └─ Tear down
├─ Build Docker Images (smoke test)
└─ RAG Evaluation (on label: "run-eval")
    └─ Ragas evaluation against golden Q&A dataset

```

7.2 CD Pipeline (on merge/tag)

```

cd-staging.yml (on merge to develop)
├─ Build + push Docker images to ECR
├─ Deploy to staging K8s cluster
└─ Run smoke tests

cd-production.yml (on tag v*)
├─ Build + push Docker images to ECR
├─ Run full Ragas evaluation
├─ Deploy to production K8s cluster (rolling update)
└─ Post-deploy health checks

```

Phase 8: RAG Evaluation — Ragas

8.1 Golden Dataset — `backend/tests/evaluation/golden_qa.json`

Hand-curated Q&A pairs from real lease documents:

```

[
  {
    "question": "Can I have pets in the apartment?",
    "ground_truth": "Pets are not allowed without prior written consent from the landlord",
    "context_pages": [3, 4]
  },
  {
    "question": "When is rent due?",
    "ground_truth": "Rent is due on the first day of each calendar month. A late fee of $50 is assessed for late payments.",
    "context_pages": [2]
  }
]

```

8.2 Evaluation Metrics

Metric	What It Measures	Target
<code>faithfulness</code>	Is the answer grounded in retrieved context?	> 0.90
<code>answer_relevancy</code>	Does the answer address the question?	> 0.85
<code>context_precision</code>	Are retrieved chunks relevant?	> 0.80
<code>context_recall</code>	Were all necessary chunks retrieved?	> 0.80

Evaluations run in CI on demand (via PR label) and on every production deploy as a gate.

Resolved Questions

Question	Decision
Authentication	None — internal tool, no auth middleware needed
Lease retention	90 days auto-cleanup (PostgreSQL <code>pg_cron</code> job deletes old records + S3 lifecycle rule)
Embedding model	<code>nomic-embed-text</code> via Ollama (768 dimensions) — best free option
Multi-lease	Single lease at a time — simpler architecture
AWS Free Tier	Architecture designed to stay within free tier (S3 5GB, Lambda 1M req, Step Functions 4K transitions, Textract 1K pages/3mo)

Verification Plan

Automated Tests

```
# Backend unit tests
cd backend && pytest tests/unit/ -v

# Backend integration tests (requires Docker services)
docker-compose -f infra/docker/docker-compose.test.yml up -d
cd backend && pytest tests/integration/ -v

# Frontend tests
cd frontend && npm test

# RAG evaluation
cd backend && pytest tests/evaluation/ -v --ragas

# Full local stack smoke test
make dev          # starts all services
make test-e2e     # runs end-to-end upload + chat test
```

Manual Verification

- Upload a real lease PDF and verify text extraction accuracy
- Upload a scanned lease image and verify OCR quality via Textract
- Ask 10+ diverse lease questions and verify answer quality + citations
- Test WebSocket reconnection by killing the backend mid-stream
- Verify Kubernetes deployment with `kind` locally before staging

Implementation Order

Phase	Scope	Estimated Effort
Phase 1	Foundation (DB, Docker, config)	2-3 days
Phase 2	Document processing pipeline (S3, Lambdas, Step Functions)	3-4 days
Phase 3	RAG & Q&A engine (LlamaIndex, LangGraph)	4-5 days
Phase 4	WebSocket chat + Redis caching	2-3 days
Phase 5	Frontend (Next.js UI)	3-4 days
Phase 6	Kubernetes manifests + deployment	2-3 days
Phase 7	CI/CD pipelines	1-2 days
Phase 8	Ragas evaluation suite	1-2 days
Total		~18-26 days